
Homework #1

Deep Learning for Computer Vision

NTU, Fall 2022

Outline

- Problem 1: Image classification
- Problem 2: Semantic segmentation
- Tools
- Submission
- Homework policy

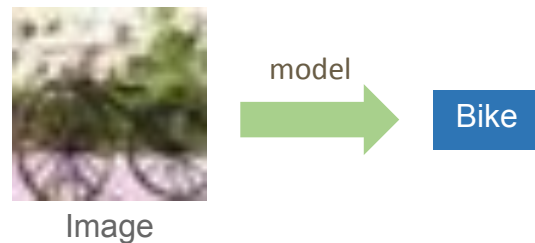
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Problem 1: Image Classification

- Image classification - predict a label for each image

- Input : RGB image
- Output : classification label



- You need to perform image classification with the following methods:
 - **A.** Train a CNN model **from scratch**
 - **B.** Try alternative models/methods (e.g., fine-tune a pre-trained model)

Dataset

- The dataset consists of 25,000 colored images (32x32 pixels) with 50 classes.
- We split the dataset into
 - **p1_data/train_50/**
 - 22500 images
 - Images are named '{class label}_{*image_id*}.png'
 - **p1_data/val_50/**
 - 2500 images
 - Naming rules are same as p1_data/train_50/
 - You can **NOT** use validation data to train your model in a fully supervised manner, but feel free to consider semi-supervised, etc. approaches using both training & validation data

Model Evaluation

- Evaluation metric: Accuracy
 - Accuracy is calculated over all test images.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Grading (45%)

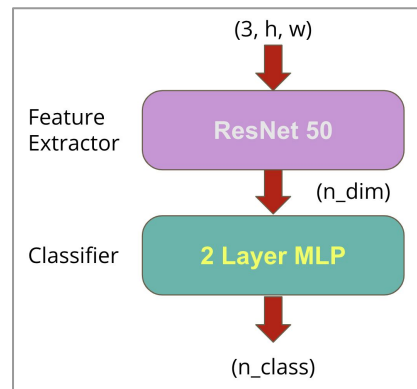
- Public Baseline (**10%**) - 2500 validation data
 - simple baseline (5%) - **0.75**
 - strong baseline (5%) - **0.86**
- Private Baseline (**10%**) - 5000 testing data
 - simple baseline (5%)
 - strong baseline (5%)
 - Will be announced after deadline
- You only need to submit one model (either A or B) for the above public/private evaluation.

Grading (45%)

Report (25%)

1. **(2%)** Draw the network architecture of method A or B.
 - The graph should be brief and clear
 - It would be fine to straight copy the figure from the paper
2. **(1%)** Report accuracy of your models (both A, B) on the validation set.
3. **(4%)** Report your implementation details of model A.
 - Including but not limited to optimizer, loss function, cross validation method
4. **(4%)** Report your alternative model or method in B, and describe its difference from model A.

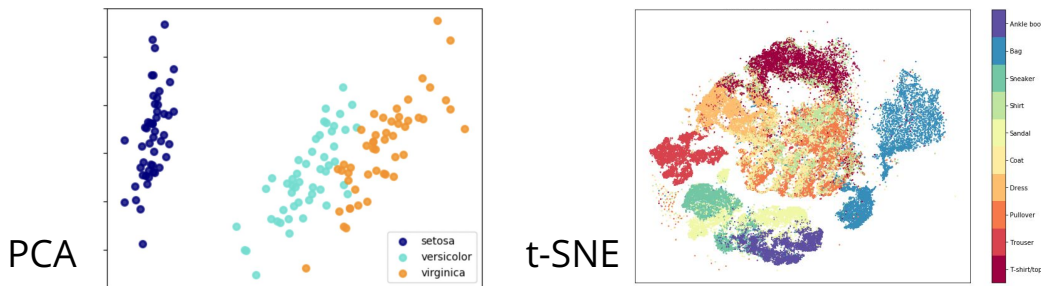
Sample Graph for Q1



Grading (45%)

Report (25%)

5. **(7%)** Visualize the learned visual representations of **model A** on the **validation set** by implementing **PCA** (Principal Component Analysis) on the output of **the second last layer**. Briefly explain your result of the PCA visualization.
6. **(7%)** Visualize the learned visual representation of **model A**, again on the output of the second last layer, but using **t-SNE** (t-distributed Stochastic Neighbor Embedding) instead. Depict your visualization from **three different epochs** including the first one and the last one. Briefly explain the above results.



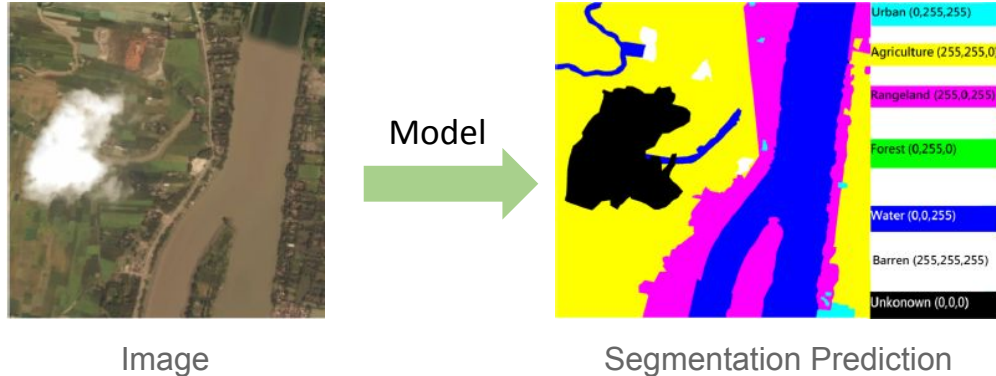
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Problem 2: Semantic Segmentation

Task Definition

- Semantic segmentation - predict the label for each pixel in an image
 - Input : RGB image
 - Output : Semantic segmentation mask



Problem 2: Semantic Segmentation

Task Definition

- You need to implement two segmentation models and provide implementation details in the report.

- **A: VGG16 + FCN32s (baseline model)**

- Implement VGG16-FCN32s model to perform segmentation.

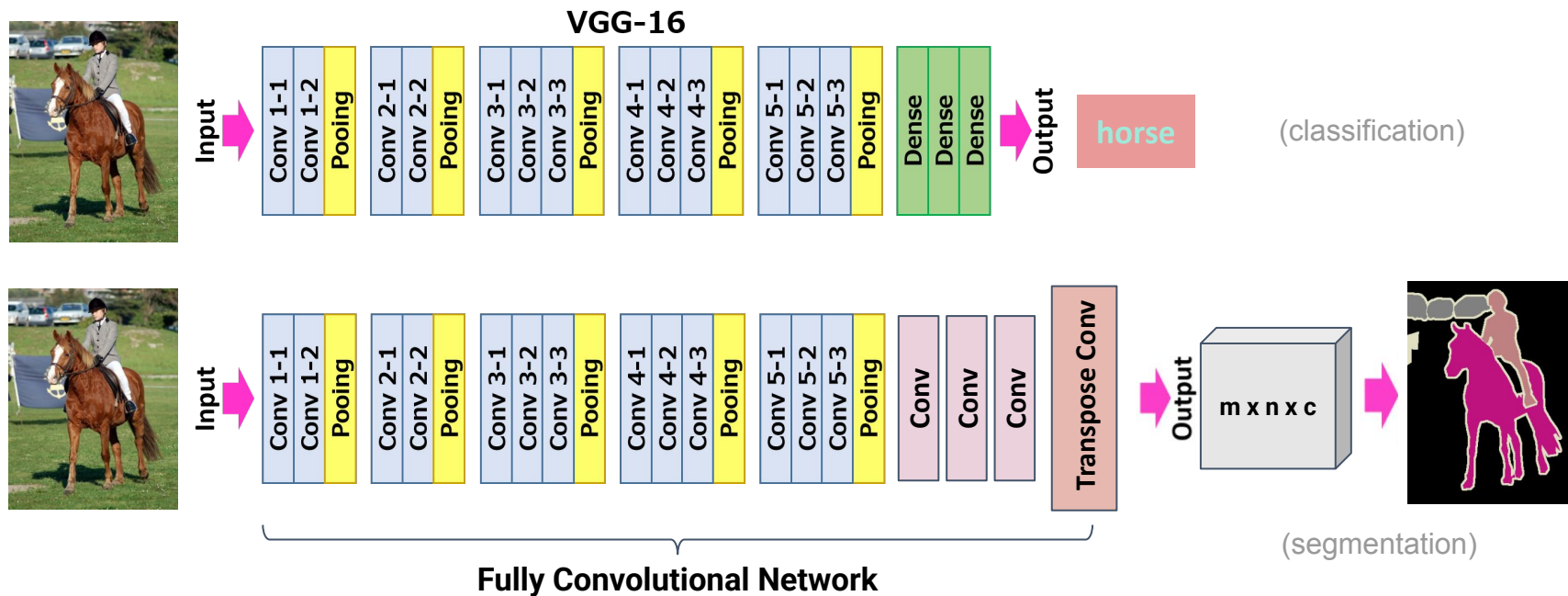
- **B: An improved model**

- Implement an improved model to perform segmentation.

- (You may choose any model different from VGG16-FCN32s, e.g., FCN16s, FCN8s, U-Net, SegNet, etc.)

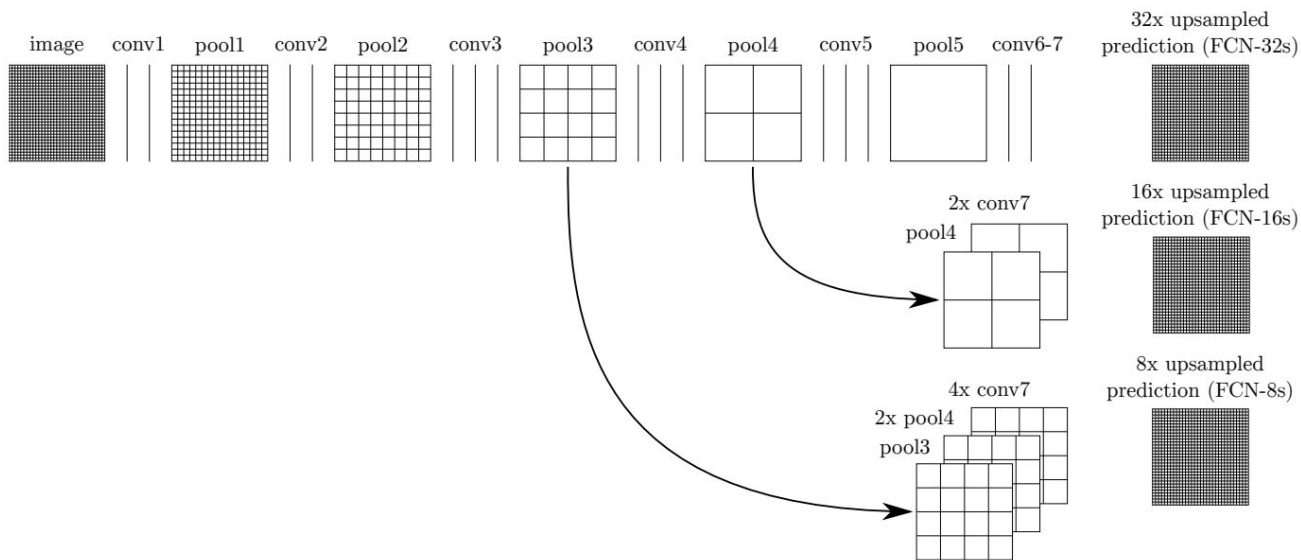
Semantic Segmentation

VGG16 + FCN32s



Semantic Segmentation

Fully Convolutional Network - FCN 32s / 16s / 8s



Dataset



Image



Semantic Segmentation Prediction

- Image size: 512x512
- Mask size: 512x512
- 7 possible class labels

- **p2_data/train/**

- Contains 2000 image-mask (ground truth) pairs
- Satellite images are named 'xxxx_sat.jpg'
- Mask images (ground truth) are named 'xxxx_mask.png'

- **p2_data/validation/**

- Contains 257 image-mask pairs
- Naming rules are the same as p2_data/train/
- You can **NOT** use validation data to train your model in a fully supervised manner, but feel free to consider semi-supervised, etc. approaches using both training & validation data

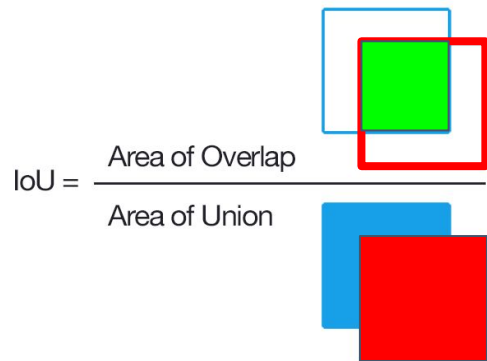
Metric

- mean Intersection over Union (mIoU)

- For each class, IoU is defined as following:

$$\text{IoU} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive} + \text{False Negative}}$$

- mean IoU is calculated by averaging over the IoU of all classes **except Unknown(0,0,0)**.
- mIoU is calculated over **all test images**.



Grading - Problem 2 (55%)

- Public Baseline (15%) - 257 validation data
 - simple baseline (5%): **0.69**
 - strong baseline (10%): **0.73**
- Private Baseline (**20%**) - 313 testing data
 - simple baseline (10%)
 - strong baseline (10%)
 - Will be annouced after deadline
- You only need to submitt one model (either A or B) for the above public/private evaluation.

Grading - Problem 2

Report (20%)

1. **(5%)** Draw the network architecture of your VGG16-FCN32s model (model A).
2. **(5%)** Draw the network architecture of the improved model (model B) and explain it differs from your VGG16-FCN32s model.
3. **(3%)** Report mIoUs of two models on the validation set.
4. **(7%)** Show the predicted segmentation mask of “validation/0013_sat.jpg”, “validation/0062_sat.jpg”, “validation/0104_sat.jpg” during the early, middle, and the final stage during the training process of the improved model.
 - Tips: Given n epochs training, you could save the 1st, $(n/2)$ -th, n -th epoch model, and draw the predicted mask by loading these saved models.

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Tool for Dataset

- **Download the dataset**

- (Option 1) Manually download the dataset here

https://drive.google.com/file/d/1LMlaOY8NSKWmGtbvTsjXcHVaYKnNN_9u

- (Option 2) Run the bash script provided in the hw1 repository

`bash ./get_dataset.sh`

Tools for Problem 2

- **mIoU**

```
python3 mean_iou_evaluate.py -g <ground_truth_directory> -p <prediction_directory>
```

- **Visualization**

```
python3 viz_mask.py <--img_path xxxx_sat.jpg> <--seg_path xxxx_mask.png>
```

PyTorch Tutorial

- We provide some resources for someone who is not familiar with **PyTorch**
 - [Introduction to PyTorch](#)
 - [Basic PyTorch classes](#)
 - [Sample colab notebook](#)

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Submission

- Deadline: **2022/10/10 (Mon.) 23:59 (GMT+8)**
- Click the following link and sign in to your GitHub account to get your submission repository:

<https://classroom.github.com/a/HbabJJJaT>

- You should connect your Github account to the classroom with your **student ID**
 - If you cannot find your student ID in the list, please contact us (ntudlcv@gmail.com)
- By default, we will only grade your latest pushed commit **before the deadline** (**NOT** your last submission).
- Please e-mail the TAs if you'd like to submit another version of your repository specifying which commit to grade.

Submission

- Your GitHub repository **DLCV-Fall-2022/hw1-{GitHub_ID}** should include the following files:
 - hw1_1.sh
 - hw1_2.sh
 - hw1_download.sh or model checkpoints
 - hw1_<studentID>.pdf (e.g. hw1_r09942249.pdf)
 - your python files (Training & Inference code, both required)
- **DO NOT push the dataset to your repo**

Shell Script (Problem 1) - hw1_1.sh

- Provide a **script** to test images under the specified directory with your model, and save the classification results in the specified csv file.
- TAs will run your script as shown below:
 - `bash hw1_1.sh $1 $2`
 - \$1: testing images directory with images named 'xxxx.png' (e.g. input/test_dir/)
 - \$2: path of output csv file (e.g. output/pred.csv)
- The output csv file **must** have the same format as 'val_gt.csv'
- This section must be finished in **10 mins**, otherwise would be considered as a failed run.

Sample CSV Format (Problem 1)

- Predict class labels for all images
 - Output format: csv file
 - The first row must be: 'filename, label'
 - You should only output the filename instead of the whole file path (i.e. Given the image path 'input/test_dir/0_450.png', you should only output '0_450.png')

```
filename,label
0_450.png,0
0_451.png,0
0_452.png,0
0_453.png,0
0_454.png,0
```

Shell Script (Problem 2) - hw1_2.sh

- Provide a **script** to test images under the specified directory with your model, and save the segmentation results as images in the specified output directory.
- TA will run your code as shown below:
 - **bash hw1_2.sh \$1 \$2**
 - \$1: testing images directory with images named 'xxxx.jpg' (e.g. input/test_dir/)
 - \$2: output images directory (e.g. output/pred_dir/)
- You **should** name your output **segmentation mask** as 'xxxx.png'
- This section must be finished in **10 mins**, otherwise would be considered as a failed run.

Model Checkpoint

- If your model checkpoints are larger than GitHub's maximum capacity (50 MB), you could download and preprocess (e.g. unzip, tar zxf, etc.) them in `hw1_download.sh`.
 - TAs will run ``bash hw1_download.sh`` prior to any inference if the download script exists, i.e. it is **NOT** necessary to create a blank ``hw1_download.sh`` file.
- Do **NOT** delete your model checkpoints before the TAs release your score and before you have ensured that your score is correct.

Download Tutorial

- Please use **wget** to download the model checkpoints from cloud drive (e.g. Dropbox) or your working station.
 - You should use **-O argument** to specify the filename of the downloaded checkpoint.
 - Please refer to this [Dropbox Guide](#) for a detailed tutorial.
- Google Drive is a widely used cloud drive, so it is allowed to use **gdown** to download your checkpoints from your drive.
 - It is also recommended to use **-O** argument to specify the filename.
 - Remember to set the permission visible to public, otherwise TAs are unable to grade your submission, resulting in zero point.
 - If you have set the permission correspondingly but failed to download with **gdown** because of Google's policy, TAs will manually download them, no worries!!

Environment

- Ubuntu 20.04.1 LTS
- NVIDIA GeForce RTX 2080 Ti (11 GB)
- GNU bash, version 5.0.17(1)-release
- Python 3.8

Packages

- imageio==2.21.2
- matplotlib==3.5.3
- numpy==1.23.1
- Pillow==9.2.0
- scipy==1.9.1
- torch==1.12.1
- torchvision==0.13.1

Optional:

- gdown, tqdm, glob, yaml
- transformers==4.21.3
- timm==0.6.7
- scikit learn==1.1.2
- pandas

If you want to use any other packages, please email TA first.

Final Check

- Ensure your code can be executed successfully on **Linux** system before your submission.
- Use only **Python3** and **Bash** script conforming to our environment, do not use other languages (e.g. CUDA) and other shell (e.g. zsh, fish) during inference.
 - Use the command ``python3`` to execute your testing python files.
- You must **NOT** use commands such as **sudo**, **CUDA_VISIBLE_DEVICES** or other commands to interfere with the environment; **any malicious attempt against the environment will lead to zero point in this assignment.**
- You shall **NOT** hardcode any path in your python files or scripts, while the dataset given would be the absolute path to the directory.

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Deadline and Academic Honesty

- Deadline: **2022/10/10 (Mon.) 23:59 (GMT+8)**
- Late policy : Late homework submission will be deducted by **30%** each day.
- **Taking any unfair advantages over other class members (or letting anyone do so) is strictly prohibited. Violating university policy would result in F for this course.**
- Students are encouraged to discuss the homework assignments, but you must complete the assignment by yourself. TA will compare the similarity of everyone's homework. **Any form of cheating or plagiarism will not be tolerated and result in an F for students with such misconduct.**

Code Modification

- If your code cannot be executed, you have a chance to make minor modifications to your code. After modifying your code,
 - If we can execute your code, you will receive a **30% penalty** in your model performance score.
 - If we still cannot execute your code, no points will be given.
- TAs will release the log of execution after grading, please check.
 - Email the TAs if something goes wrong in your submission.

How to find help

- Google!
- Use TA hours (please check [course website](#) for time/location)
- Post your question under hw1 FAQ section in FB group for discussion
- Post your question to NTU COOL
- Contact TAs by e-mail: ntudlcv@gmail.com
- Find the easter eggs in the slide!

Dos and Dont's for the TAs

- Do NOT send private messages to TAs via Facebook or spam TAs email.
 - TAs are happy to help, but they are not your tutors 24/7.
- TAs will NOT debug for you (e.g., coding, environment, dependencies, etc. issues).
- If you cannot make the TA hours, please email the TAs for an appointment.

Final Reminder

- Please start working on this homework **as early as possible**.

The training may take a few hours on a GPU or days on CPUs.

- As mentioned in the first class, we are NOT responsible for any computation resources.
- Failure to conform to the aforementioned regulations will incur penalty.
- TAs are here to help. Discussions/Q&As are welcomed :)